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Display device and operation method for heating display panel

Abstract

The disclosure provides a display device and an operation method thereof. The display device includes a display panel and a driving circuit. The display panel has at least one native conductive line for a display driving operation. The driving circuit is coupled to the at least one native conductive line. During a display driving period, the driving circuit performs the display driving operation on the display panel through the at least one native conductive line. During a heating period, the driving circuit performs a heating operation on the display panel through the at least one native conductive line.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8654117	12/2013	Yanase et al.	N/A	N/A
9176338	12/2014	Yang et al.	N/A	N/A
9470933	12/2015	Yanase et al.	N/A	N/A
10795199	12/2019	Zhao et al.	N/A	N/A
2001/0043166	12/2000	Jacobsen	348/E5.143	H04M 1/021
2006/0071890	12/2005	Kikuchi	345/87	G09G 3/3677
2007/0018919	12/2006	Zavracky	345/87	G09G 3/3413
2007/0018923	12/2006	Yokota	345/89	G09G 3/3688
2009/0256820	12/2008	Yanase et al.	N/A	N/A
2014/0132859	12/2013	Yanase et al.	N/A	N/A
2016/0291782	12/2015	Zhai	N/A	G06F 3/04164
2018/0240932	12/2017	Chen	N/A	N/A
2020/0058533	12/2019	Zou et al.	N/A	N/A
2021/0200354	12/2020	Kang	N/A	G06F 3/0416
2021/0383755	12/2020	Kawachi	N/A	G09G 3/3233
2023/0178411	12/2022	Zhao et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101510020	12/2009	CN	N/A
103336593	12/2012	CN	N/A
106903978	12/2018	CN	N/A
M595307	12/2019	TW	N/A

OTHER PUBLICATIONS

“Office Action of Taiwan Counterpart Application”, issued on Aug. 8, 2024, p. 1-p. 4. cited by applicant

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Background/Summary

BACKGROUND

Technical Field

(1) The disclosure relates to an electronic device, and particularly relates to a display device and an operation method thereof.

Description of Related Art

(2) The operating temperature range of a general liquid-crystal display (LCD) panel is approximately -20°C . to 70°C . At low temperatures (less than -20°C .), liquid crystals rotate very slowly. When the screen on the LCD panel starts to change, image retention appears on the screen due to the slow rotation speed of the liquid crystals. Therefore, how to heat the LCD panel in a low temperature environment is one of many technical issues in this field.

(3) There are three existing LCD panel heating technologies. The first method is the plug-in type, that is, to use an additional heating equipment to increase the temperature of the LCD panel. The second method is to coat an additional metal heating layer on the upper layer of the LCD panel, and then use this additional metal heating layer to increase the temperature of the LCD panel. The third method is to use an extra metal conductive line as a heating conductive line in the LCD panel, and then use this extra metal conductive line to increase the temperature of the LCD panel. The above three methods all require additional design or modifying the original manufacturing operations, which increases the cost. In addition, if the LCD panel has a touch detection function, the first method and the second method may block the touch signal. The existing LCD panel heating technologies are not conducive to the current LCD in-cell touch panel.

SUMMARY

(4) The disclosure provides a display device and an operation method thereof to heat a display panel.

(5) In an embodiment of the disclosure, the display device includes the display panel and a driving circuit. The display panel has at least one native conductive line for a display driving operation. The driving circuit is coupled to the at least one native conductive line. During a display driving period, the driving circuit performs the display driving operation on the display panel through the at least one native conductive line. During a heating period, the driving circuit performs a heating operation on the display panel through the at least one native conductive line.

(6) In an embodiment of the disclosure, the operation method includes the following. During a display driving period, a display driving operation is performed on a display panel by a driving circuit of a display device through at least one native conductive line of the display panel. During a heating period, a heating operation is performed on the display panel by the driving circuit through the at least one native conductive line.

(7) Based on the above, the native conductive line according to the various embodiments is not merely configured to perform the display driving operation during the display driving period, but can also be configured to perform the heating operation during the heating period. Therefore, the display device can increase the temperature of the display panel without using an additional heating device.

(8) In order to make the above-mentioned features and advantages of the disclosure more comprehensible, the embodiments are described in detail below with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic circuit block diagram of a display device according to an embodiment of the disclosure.

(2) FIG. 2 is a schematic flowchart of an operation method of the display device according to an embodiment of the disclosure.

- (3) FIG. 3 is a schematic circuit block diagram of a driving circuit and a display panel according to an embodiment of the disclosure.
- (4) FIG. 4 is a schematic driving waveform diagram of a gate driver performing a driving operation on a gate line according to an embodiment of the disclosure.
- (5) FIG. 5 is a schematic driving waveform diagram of a gate driver performing a driving operation on a gate line according to another embodiment of the disclosure.
- (6) FIG. 6 is a schematic circuit block diagram of a driving circuit according to another embodiment of the disclosure.
- (7) FIG. 7 is a schematic circuit block diagram of a driving circuit according to still another embodiment of the disclosure.
- (8) FIG. 8 is a schematic circuit block diagram of a driving circuit according to yet another embodiment of the disclosure.
- (9) FIG. 9 is a schematic driving waveform diagram of a source driver performing a driving operation on a source line according to an embodiment of the disclosure.
- (10) FIG. 10 is a schematic circuit block diagram of a driving circuit according to further another embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(11) The term “coupled (or connected)” used throughout the specification of the disclosure (including the appended claims) may refer to any direct or indirect means of connection. For example, if it is described that the first device is coupled (or connected) to the second device, it should be interpreted that the first device may be directly connected to the second device, or the first device may be connected indirectly to the second device through other devices or some kind of connection means. The terms “first” and “second” mentioned throughout the specification of the disclosure (including the appended claims) are used to name the name of an element, or to distinguish different embodiments or ranges, and are not used to limit the upper limit or the lower limit of the number of elements, and are not to limit the order of elements either. In addition, wherever possible, elements/members/steps using the same reference numerals in the drawings and embodiments represent the same or similar parts. For the elements/members/steps using the same reference numerals or using the same terms, reference may be made to the related descriptions in different embodiments.

(12) FIG. 1 is a schematic circuit block diagram of a display device **100** according to an embodiment of the disclosure. The display device **100** shown in FIG. 1 includes a driving circuit **110** and a display panel **120**. This embodiment does not limit the implementation details of the display panel **120**. Based on the actual design, the display panel **120** may include a liquid-crystal display (LCD) panel or other display panels. The display panel **120** has one or more native conductive lines (e.g., a native conductive line NL1 shown in FIG. 1) for performing a display driving operation. For example, the native conductive line may include a gate line (or referred to as a scan line), a source line (or referred to as a data line), and/or other transmission lines. These native conductive lines are configured to perform the display driving operation. In addition, some (or all) of these native conductive lines can also be configured to perform a heating operation.

(13) FIG. 2 is a schematic flowchart of an operation method of the display device according to an embodiment of the disclosure. Please refer to FIG. 1 and FIG. 2. The driving circuit **110** is coupled to one or more native conductive lines (e.g., the native conductive line NL1) of the display panel **120**. During a display driving period, the driving circuit **110** may perform the display driving operation on the display panel **120** through the native conductive line (Step S210). This embodiment does not limit the implementation details of the display driving operation. Based on the actual design, the display driving operation may include a well-known display driving operation or other display driving operations.

(14) During a heating period, the driving circuit **110** may perform the heating operation on the display panel **120** through the native conductive line (Step S220). For example, the driving circuit

110 may apply different voltages to two terminals of the native conductive line **NL1**, that is, apply a certain heating voltage difference ΔV to the two terminals of the native conductive line **NL1**, so that a heating current passes through the native conductive line **NL1**. Since the native conductive line **NL1** has a parasitic resistance, the heating current can make the native conductive line **NL1** heat up. The heat generation characteristic of the native conductive line **NL1** is shown in the following Formula (1). Q is the heat emitted by the native conductive line **NL1**, P is the power consumed by the native conductive line **NL1**, T is time, ΔV is the heating voltage difference applied to the two terminals of the native conductive line **NL1**, R is the parasitic resistance value of the native conductive line **NL1**. It may be seen from the Formula (1) that when there is a voltage difference between the two terminals of the native conductive line **NL1**, the native conductive line **NL1** can generate heat, thereby heating up the display panel **120**.

(15) $Q = P \times T = [(\Delta V^2 / R) \times T]$ Formula(1)

(16) Based on the above, the driving circuit **110** described in this embodiment can perform the display driving operation during the display driving period by using the native conductive line **NL1**. This embodiment does not limit the actual implementation of the display driving period. For example, based on the actual design, the display driving period may be a well-known display driving period. In addition, the driving circuit **110** may also perform the heating operation during the heating period by using the native conductive line **NL1**. This embodiment does not limit the actual implementation of the heating period. For example, based on the actual design, the heating period may be outside the display driving period. In some actual design examples, the heating period may be within a vertical porch period (a period that is not of a display operation or a touch operation) or a vertical blanking period. Therefore, the display device **100** can increase the temperature of the display panel **120** without using an additional heating device.

(17) FIG. 3 is a schematic circuit block diagram of the driving circuit **110** and the display panel **120** according to an embodiment of the disclosure. The driving circuit **110** and the display panel **120** shown in FIG. 3 may be used as one of many implementation examples of the driving circuit **110** and the display panel **120** shown in FIG. 1. In the embodiment shown in FIG. 3, the native conductive line of the display panel **120** may include multiple gate lines (or referred to as scan lines, such as gate lines **G1**, **G2**, . . . , **Gm** shown in FIG. 3) and multiple source lines (or referred to as data lines, such as source lines **S1**, **S2**, . . . , **Sn** shown in FIG. 3). The driving circuit **110** may include a gate driver **310**, a gate driver **320**, and a source driver **330**. The source driver **330** may have a timing controller function. Based on the timing control of the source driver **330**, the gate drivers **310** and **320** may jointly perform a scanning operation in the display driving operation on the gate lines **G1** to **Gm** during the display driving period. Based on the scan timing of the gate drivers **310** and **320**, the source driver **330** may perform a gray scale driving operation in the display driving operation on the source lines **S1** to **Sn** during the display driving period, so as to write a gray scale voltage into different pixel circuits **P** of the display panel **120**.

(18) FIG. 4 is a schematic driving waveform diagram of the gate drivers **310** and **320** performing the driving operation on the gate lines **G1** to **Gm** according to an embodiment of the disclosure. The horizontal axis in FIG. 4 represents time. Please refer to FIG. 3 and FIG. 4. The upper part of FIG. 4 shows the driving waveform of the driving operation performed by the gate driver **310** on terminals of the gate lines **G1** to **Gm**, and the lower part of FIG. 4 shows the driving waveform of the driving operation performed by the gate driver **320** on the other terminals of the gate lines **G1** to **Gm**.

(19) The gate driver **310** is coupled to terminals of the gate lines **G1** to **Gm** (e.g., first terminals). The gate driver **320** is coupled to the other terminals of the gate lines **G1** to **Gm** (e.g., second terminals). The gate drivers **310** and **320** may jointly perform the scanning operation on the gate lines **G1** to **Gm** during a display driving period **DP**, as shown in FIG. 4. In the embodiment shown in FIG. 4, the heating period may be within a vertical porch period **Vporch**. During the vertical

porch period V_{porch} (the heating period), the gate drivers **310** and **320** may jointly apply the certain heating voltage difference ΔV to the two terminals of each (the native conductive line NL1) of the gate lines G1 to G_m to perform the heating operation on the display panel **120**. During the heating period, the voltage applied by the gate drivers **310** and **320** to the gate lines G1 to G_m in order to heat the display panel **120** is not enough to turn on any pixel circuit P coupled to the gate lines G1 to G_m.

(20) For example (but not limited thereto), the gate driver **310** pulls down the voltages at the first terminals of the gate lines G1 to G_m from a voltage level VG1 to a voltage level VG2 during the vertical porch period V_{porch}, and the gate driver **320** maintains voltages of the second terminals of the gate lines G1 to G_m at the voltage level VG1 during the vertical porch period V_{porch}.

Therefore, the two terminals of each of the gate lines G1 to G_m have the heating voltage difference $\Delta V = VG1 - VG2$. The gate drivers **310** and **320** apply the heating voltage difference ΔV to the two terminals of each of the gate lines G1 to G_m, so that the heating current passes through the gate lines G1 to G_m. Since the gate line has the parasitic resistance, the heating current can make the gate lines G1 to G_m heat up. Therefore, the gate drivers **310** and **320** can perform the heating operation on the display panel **120** through the gate lines G1 to G_m during the heating period.

(21) FIG. 5 is a schematic driving waveform diagram of the gate drivers **310** and **320** performing the driving operation on the gate lines G1 to G_m according to another embodiment of the disclosure. The horizontal axis in FIG. 5 represents time. In the embodiment shown in FIG. 5, the driving circuit **110** also performs a touch detection operation on the display panel **120** through the one or more native conductive lines during a touch detection period TP. Please refer to FIG. 3 and FIG. 5. The gate driver **310** is coupled to the terminals of the gate lines G1 to G_m (e.g., the first terminals). The gate driver **320** is coupled to the other terminals of the gate lines G1 to G_m (e.g., the second terminals). The upper part of FIG. 5 shows the driving waveform of the driving operation performed by the gate driver **310** on the first terminals of the gate lines G1 to G_m, and the lower part of FIG. 5 shows the driving waveform of the driving operation performed by the gate driver **320** on the second terminals of the gate lines G1 to G_m. The gate drivers **310** and **320** may jointly perform the scanning operation on the gate lines G1 to G_m during the display driving period DP, and jointly perform LFD (Load Free Driving, i.e. the touch detection operation) on the display panel **120** through the gate lines G1 to G_m during the touch detection period TP, as shown in FIG. 5.

(22) In the embodiment shown in FIG. 5, the heating period may be outside the display driving period DP and the touch detection period TP. For example, the heating period may be within the vertical porch period V_{porch} or the vertical blanking period. During the vertical porch period V_{porch} (the heating period), the gate driver **310** and **320** may jointly apply the certain heating voltage difference ΔV to the two terminals of each of the gate lines G1 to G_m (the native conductive line NL1) to perform the heating operation on the display panel **120**. For the heating operation of the vertical porch period V_{porch} shown in FIG. 5, reference may be made to the relevant description of the heating operation of the vertical porch period V_{porch} shown in FIG. 4, so details are not repeated here.

(23) FIG. 6 is a schematic circuit block diagram of the driving circuit **110** according to another embodiment of the disclosure. For the display panel **120** shown in FIG. 6, reference may be made to the relevant description of the display panel **120** shown in FIG. 3, so details are not repeated here. The driving circuit **110** shown in FIG. 6 may be used as one of many implementation examples of the driving circuit **110** shown in FIG. 1. In the embodiment shown in FIG. 6, the driving circuit **110** may include a gate driver **610**, a gate driver **620**, and a source driver **630**. For the source driver **630** shown in FIG. 6, reference may be made to the relevant description of the source driver **330** shown in FIG. 3, for the operation of the gate drivers **610** and **620** during the display driving period shown in FIG. 6, reference may be made to the relevant descriptions of the gate drivers **310** and **320** shown in FIG. 3, so details are not repeated here. Based on the timing

control of the source driver **630**, the gate drivers **610** and **620** may jointly perform the scanning operation on the gate lines **G1** to **Gm** during the display driving period. Based on the scan timing of the gate drivers **610** and **620**, the source driver **630** may perform the gray scale driving operation on the source lines **S1** to **Sn** during the display driving period, so as to write the gray scale voltage into the different pixel circuits **P** of the display panel **120**.

(24) In the embodiment shown in FIG. **6**, the gate driver **620** may include a scanning circuit **621** and switch circuits **622_1** to **622_i**. For other switch circuits (such as the switch circuit **622_i**) in the gate driver **620**, reference may be made to the relevant description of the switch circuit **622_1** and make analogies. In the embodiment shown in FIG. **6**, the switch circuit **622_1** includes a switch **SW1**, a switch **SW2**, and a switch **SW3**. During the display driving period, the switch **SW1** and the switch **SW2** are turned on and the switch **SW3** is turned off, so the gate driver **610** and the scanning circuit **621** may jointly perform the scanning operation on the gate lines **G1** and **G2**.

(25) During the heating period, the switch **SW1** and the switch **SW2** are turned off and the switch **SW3** is turned on, so the gate driver **620** may couple the second terminal of the gate line **G1** to the second terminal of the gate line **G2**. During the heating period, the gate driver **610** may apply the heating voltage difference ΔV to the first terminal of the gate line **G1** and the first terminal of the gate line **G2**, in which the voltage applied by the gate driver **610** to the gate lines **G1** and **G2** in order to heat the display panel **120** during the heating period is not enough to turn on the pixel circuit **P** coupled to the gate lines **G1** and **G2**. For example (but not limited thereto), the gate driver **310** pulls down the voltage at the first terminal of the gate line **G1** from the voltage level **VG1** to the voltage level **VG2** during the heating period (such as the vertical porch period **Vporch**), and maintains the voltage of the first terminal of the gate line **G2** at the voltage level **VG1**. Therefore, the heating current passes through the gate lines **G1** and **G2**. Since the gate line has the parasitic resistance, the heating current can make the gate lines **G1** and **G2** heat up, thereby performing the heating operation on the display panel **120**. In the same way, by adding a switch circuit into the gate driver **610**, the switch circuit can connect three or more gate lines during the heating period to heat the display panel **120**.

(26) FIG. **7** is a schematic circuit block diagram of the driving circuit **110** according to still another embodiment of the disclosure. For the display panel **120** shown in FIG. **7**, reference may be made to the relevant description of the display panel **120** shown in FIG. **3**, so details are not repeated here. The driving circuit **110** shown in FIG. **7** may be used as one of many implementation examples of the driving circuit **110** shown in FIG. **1**. In the embodiment shown in FIG. **7**, the driving circuit **110** may include a gate driver **710**, a gate driver **720**, and a source driver **730**. For the source driver **730** shown in FIG. **7**, reference may be made to the relevant description of the source driver **330** shown in FIG. **3**, so details are not repeated here. The operation of the gate drivers **710** and **720** on the gate lines **G1** and **G2** will be described below with specific examples. For operations of the gate driver **710** and **720** on other gate lines of the display panel **120**, reference may be made to the relevant descriptions of the gate lines **G1** and **G2** and make analogies.

(27) The gate driver **710** is coupled to the first terminal of the gate line **G1** and the first terminal of the gate line **G2**, and the gate driver **720** is coupled to the second terminal of the gate line **G1** and the second terminal of the gate line **G2**. In the embodiment shown in FIG. **7**, the gate driver **710** includes a scanning circuit **711** and a switch **SW4**, and the gate driver **720** includes a scanning circuit **721** and a switch **SW5**. The scanning circuit **711** is coupled to the first terminal of the gate line **G1**. The first terminal of the switch **SW4** receives the voltage level **VG1**. The second terminal of the switch **SW4** is coupled to the first terminal of the gate line **G2**. The first terminal of the switch **SW5** receives the voltage level **VG1**. The second terminal of the switch **SW5** is coupled to the second terminal of the gate line **G1**. The scanning circuit **721** is coupled to the second terminal of the gate line **G2**.

(28) During the display driving period, the switch **SW4** and the switch **SW5** are turned off. Based on the timing control of the source driver **730**, during the display driving period, the scanning

circuit **711** of the gate driver **710** may perform a first scanning operation on an odd-numbered gate line (such as the gate line **G1**) among the gate lines **G1** to **Gm**, and the scanning circuit **721** of the gate driver **720** may perform a second scanning operation on an even-numbered gate line (such as the gate line **G2**) among the gate lines **G1** to **Gm**. Based on the scan timing of the scanning circuits **711** and **721**, the source driver **730** may perform the gray scale driving operation on the source lines **S1** to **Sn** during the display driving period, so as to write the gray scale voltage into the different pixel circuits **P** of the display panel **120**.

(29) During the heating period (such as the vertical porch period **Vporch**), the switch **SW4** and the switch **SW5** are turned on, so the voltage level **VG1** is applied to the second terminal of the gate line **G1** and the first terminal of the gate line **G2**. The scanning circuit **711** may pull down the voltage at the first terminal of the gate line **G1** from the voltage level **VG1** to the voltage level **VG2** during the heating period, so the gate drivers **710** and **720** jointly apply the heating voltage difference ΔV to the two terminals of the gate line **G1** to perform the heating operation on the display panel **120**. In addition, the scanning circuit **721** may pull down the voltage at the second terminal of the gate line **G2** from the voltage level **VG1** to the voltage level **VG2** during the heating period, so the gate drivers **710** and **720** jointly apply the heating voltage difference ΔV to the two terminals of the gate line **G2** to perform the heating operation on the display panel **120**. During the heating period, the voltage applied to the gate line **G1** and the voltage applied to the gate line **G2** by the gate drivers **710** and **720** in order to heat the display panel **120** are not enough to turn on the pixel circuits **P** coupled to the gate lines **G1** and **G2**.

(30) FIG. **8** is a schematic circuit block diagram of the driving circuit **110** according to yet another embodiment of the disclosure. For the display panel **120** shown in FIG. **8**, reference may be made to the related description of the display panel **120** shown in FIG. **3**, so details are not repeated here. The driving circuit **110** shown in FIG. **8** may be used as one of many implementation examples of the driving circuit **110** shown in FIG. **1**. In the embodiment shown in FIG. **8**, the driving circuit **110** may include a gate driver **810**, a gate driver **820**, a source driver **830**, and a plurality of switches (such as switches **840_1** to **840_j** as shown in FIG. **8**). For the display driving operation of the gate driver **810**, the gate driver **820**, and the source driver **830** on the display panel **120** shown in FIG. **8**, reference may be made to the relevant descriptions of the gate driver **310**, the gate driver **320**, and the source driver **330** shown in FIG. **3**, so details are not repeated here. Based on the timing control of the source driver **830**, the gate drivers **810** and **820** may jointly perform the scanning operation on the gate lines **G1** to **Gm** during the display driving period. Based on the scan timing of the gate drivers **810** and **820**, the source driver **830** may perform the gray scale driving operation on the source lines **S1** to **Sn** during the display driving period, so as to write the gray scale voltage into the different pixel circuits **P** of the display panel **120**.

(31) FIG. **9** is a schematic driving waveform diagram of the source driver **830** performing the driving operation on the source lines **S1** and **S2** according to an embodiment of the disclosure. The horizontal axis in FIG. **9** represents time. Please refer to FIG. **8** and FIG. **9**. For other switches in the driving circuit **110** (such as the switch **840_j**), reference may be made to the relevant description of the switch **840_1** and make analogies, and for other source lines of the display panel **120** (such as the source line **Sn**), reference may be made to the relevant descriptions of the source lines **S1** and **S2** and make analogies. The source driver **830** is coupled to the first terminal of the source line **S1** and the first terminal of the source line **S2**. The first terminal and the second terminal of the switch **840_1** are respectively coupled to the second terminal of the source line **S1** and the second terminal of the source line **S2**. During the display driving period **DP**, the switch **840_1** is turned off, so based on the scan timing of the gate drivers **810** and **820**, the source driver **830** may perform the gray scale driving operation on the source lines **S1** and **S2**, so as to write the gray scale voltage into the different pixel circuits **P** of the display panel **120**.

(32) In the embodiment shown in FIG. **9**, the heating period may be within the vertical porch period **Vporch**. During the vertical porch period **Vporch** (the heating period), the switch **840_1** is

turned on, so that the second terminal of the source line S1 is coupled to the second terminal of the source line S2. At this time, the source driver 830 may apply the certain heating voltage difference ΔV to the first terminal of the source line S1 and the first terminal of the source line S2 to perform the heating operation on the display panel 120.

(33) For example (but not limited thereto), the source driver 830 pulls up the voltage of the first terminal of the source line S1 from the voltage level VG2 to the voltage level VG1 during the vertical porch period Vporch, and the source driver 830 maintains the voltage of the first terminal of the source line S2 at the voltage level VG2 during the vertical porch period Vporch. Therefore, the source lines S1 and S2 have the heating voltage difference $\Delta V = VG1 - VG2$, so that the heating current passes through the source lines S1 and S2. Since the source line has the parasitic resistance, the heating current can make the source lines S1 and S2 heat up. Therefore, the source driver 830 may perform the heating operation on the display panel 120 through the source lines S1 and S2 (the native conductive line NL1) during the heating period.

(34) FIG. 10 is a schematic circuit block diagram of the driving circuit 110 according to further another embodiment of the disclosure. For the display panel 120 shown in FIG. 10, reference may be made to the relevant description of the display panel 120 shown in FIG. 3, so details are not repeated here. The driving circuit 110 shown in FIG. 10 may be used as one of many implementation examples of the driving circuit 110 shown in FIG. 1. In the embodiment shown in FIG. 10, the driving circuit 110 may include a gate driver 1010, a gate driver 1020, a source driver 1030, and a plurality of switches (such as switches 1040_1, 1040_2, and 1040_3 as shown in FIG. 8). For the display driving operation of the gate driver 1010, the gate driver 1020, and the source driver 1030 on the display panel 120 shown in FIG. 10, reference may be made to the relevant descriptions of the gate driver 310, the gate driver 320, and the source driver 330 shown in FIG. 3, or reference may be made to the relevant descriptions of the gate driver 810, the gate driver 820, and the source driver 830 shown in FIG. 8, so details are not repeated here. Based on the timing control of the source driver 1030, the gate drivers 1010 and 1020 may jointly perform the scanning operation on the gate lines G1 to Gm during the display driving period DP. Based on the scan timing of the gate drivers 1010 and 1020, the source driver 1030 may perform the gray scale driving operation on the source lines S1 to S4 during the display driving period DP, so as to write the gray scale voltage into the different pixel circuits P of the display panel 120.

(35) The source driver 1030 is coupled to the first terminal of the source line S1, the first terminal of the source line S2, the first terminal of the source line S3, and the first terminal of the source line S4. The first terminal and the second terminal of the switch 1040_1 are respectively coupled to the second terminal of the source line S1 and the second terminal of the source line S2. The first terminal and the second terminal of the switch 1040_2 are respectively coupled to the first terminal of the source line S2 and the first terminal of the source line S3. The first terminal and the second terminal of the switch 1040_3 are respectively coupled to the second terminal of the source line S3 and the second terminal of the source line S4. During the display driving period DP, the switches 1040_1, 1040_2, and 1040_3 are turned off. At this time, the source driver 1030 may perform the gray scale driving operation on the source lines S1, S2, S3, and S4, so as to write the gray scale voltage into the different pixel circuits P of the display panel 120.

(36) During the heating period (such as the vertical porch period Vporch), the switch 1040_1 is turned on to couple the second terminal of the source line S1 to the second terminal of the source line S2, and the switch 1040_2 is turned on to couple the first terminal of the source line S2 to the first terminal of the source line S3, and the switch 1040_3 is turned on to couple the second terminal of the source line S3 to the second terminal of the source line S4. At this time, the source driver 1030 may apply the certain heating voltage difference ΔV to the first terminal of the source line S1 and the first terminal of the source line S4 to perform the heating operation on the display panel 120. In the same way, by adding more switches 1040 into the driving circuit 110, the switches 1040 can connect five or more source lines during the heating period to heat the display panel 120.

(37) In summary, the native conductive line NL1 according to the various embodiments is not merely configured to perform the display driving operation during the display driving period DP, but can also be configured to perform the heating operation during the heating period (such as the vertical porch period Vporch). Therefore, the display device 100 can increase the temperature of the display panel 120 without using an additional heating equipment.

(38) Although the disclosure has been disclosed above as the embodiments, the embodiments are not used to limit the disclosure. Persons with ordinary knowledge in the technical field may make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the appended claims.

Claims

1. A display device, comprising: a display panel having at least one native conductive line for performing a display driving operation; and a driving circuit coupled to the at least one native conductive line, wherein the driving circuit performs the display driving operation on the display panel through the at least one native conductive line during a display driving period, and the driving circuit performs a heating operation on the display panel through the at least one native conductive line during a heating period, wherein the driving circuit also performs a touch detection operation on the display panel through the at least one native conductive line during a touch detection period, wherein the heating period is within a vertical porch period outside the display driving period and the touch detection period, wherein the at least one native conductive line comprises a first source line, a second source line, a third source line, and a fourth source line of the display panel, and the driving circuit comprises: a source driver coupled to a first terminal of the first source line, a first terminal of the second source line, a first terminal of the third source line, and a first terminal of the fourth source line, wherein, during the display driving period, the source driver performs a gray scale driving operation of the display driving operation on the first source line, the second source line, the third source line, and the fourth source line; and during the heating period, a second terminal of the first source line is coupled to a second terminal of the second source line, the first terminal of the second source line is coupled to the first terminal of the third source line, a second terminal of the third source line is coupled to a second terminal of the fourth source line, and the source driver applies a heating voltage difference to the first terminal of the first source line and the first terminal of the fourth source line to perform the heating operation on the display panel.

2. The display device as claimed in claim 1, wherein the driving circuit further comprises: a first switch having a first terminal and a second terminal respectively coupled to a second terminal of the first source line and a second terminal of the second source line; a second switch having a first terminal and a second terminal respectively coupled to the first terminal of the second source line and the first terminal of the third source line; and a third switch having a first terminal and a second terminal respectively coupled to a second terminal of the third source line and a second terminal of the fourth source line, wherein, during the display driving period, the first switch, the second switch, and the third switch are turned off; and during the heating period, the first switch is turned on, so that the second terminal of the first source line is coupled to the second terminal of the second source line, the second switch is turned on, so that the first terminal of the second source line is coupled to the first terminal of the third source line, and the third switch is turned on, so that the second terminal of the third source line is coupled to the second terminal of the fourth source line.

3. An operation method of a display device, comprising: performing a display driving operation on the display panel through at least one native conductive line of a display panel of the display device during a display driving period by a driving circuit of the display device; performing a heating operation on the display panel through the at least one native conductive line during a heating

period by the driving circuit; and performing a touch detection operation on the display panel through the at least one native conductive line during a touch detection period by the driving circuit, wherein the heating period is within a vertical porch period outside the display driving period and the touch detection period, wherein the at least one native conductive line comprises a first source line, a second source line, a third source line, and a fourth source line of the display panel, and the driving circuit comprises a source driver coupled to a first terminal of the first source line, a first terminal of the second source line, a first terminal of the third source line, and a first terminal of the fourth source line; performing a gray scale driving operation of the display driving operation on the first source line, the second source line, the third source line, and the fourth source line by the source driver during the display driving period; during the heating period, coupling a second terminal of the first source line to a second terminal of the second source line, coupling the first terminal of the second source line to the first terminal of the third source line, coupling a second terminal of the third source line to a second terminal of the fourth source line; and applying a heating voltage difference to the first terminal of the first source line and the first terminal of the fourth source line to perform the heating operation on the display panel by the source driver during the heating period.

4. The operation method as claimed in claim 3, wherein the driving circuit further comprises a first switch, a second switch, and a third switch, a first terminal and a second terminal of the first switch are respectively coupled to a second terminal of the first source line and a second terminal of the second source line, a first terminal and a second terminal of the second switch are respectively coupled to the first terminal of the second source line and the first terminal of the third source line, a first terminal and a second terminal of the third switch are respectively coupled to a second terminal of the third source line and a second terminal of the fourth source line, and the operation method further comprises: turning off the first switch, the second switch, and the third switch during the display driving period; turning on the first switch to couple the second terminal of the first source line to the second terminal of the second source line during the heating period; turning on the second switch to couple the first terminal of the second source line to the first terminal of the third source line during the heating period; and turning on the third switch to couple the second terminal of the third source line to the second terminal of the fourth source line during the heating period.
